

# Kasteleyn's Theorem

James Bona-Landry

## 1 Review & Motivation

In this talk, we return to our study of the Dimer model; which, if we recall, is concerned with the set of *perfect matchings* (or *dimer configurations*) of graphs. In particular, we've seen that a dimer model  $(G, \mathcal{D}, \nu)$  on a graph  $G$  induces a probability measure on the set  $\mathcal{D}$  of dimer configurations of  $G$  via

$$\mathbb{P}(D) = \frac{\nu(D)}{Z(G; \nu)}, \quad \text{where } Z(G; \nu) = \prod_{D \in \mathcal{D}} \nu(D) = \prod_{D \in \mathcal{D}} \sum_{e \in D} \nu(e).$$

Recall that the normalizing constant  $Z = Z(G; \nu)$  has a special name: the *partition function*. We've seen that in certain simplistic cases,  $Z$  can be computed exactly, and hence so too can the law of  $\mathbb{P}$ . By employing the use of *height functions*, this gave us a way of generating random discrete surfaces, using the underlying graph  $G$  as a blueprint. The purpose of this talk is to vastly expand our repertoire of valid blueprints by proving the following 1961 result of Kasteleyn:

**Theorem** (Kasteleyn's Theorem). *If  $G = (V, E)$  is any finite planar graph, then for any dimer model on  $G$  there exists an orientation  $K$  of the edges of  $G$  such that  $Z = \sqrt{\det A}$ , where  $A$  denotes the weighted adjacency matrix associated to  $K$ .*

Recall that an orientation of the edges of a graph is simply a choice of direction for each edge. Recall, moreover, that the weighted adjacency matrix associated to such an orientation is the skew-symmetric  $|V| \times |V|$  matrix with  $(i, j)$ -entry given by

$$a_{ij} = \sum_e \varepsilon_{ij}(e) \nu(e), \quad \varepsilon_{ij}(e) = \begin{cases} 1 & \text{if } e: i \rightarrow j \\ -1 & \text{if } e: j \rightarrow i \end{cases}$$

where summation extends over all edges  $e$  between vertices  $i$  and  $j$ , and the notation  $e: i \rightarrow j$ , for instance, reads “ $e$  is oriented from  $i$  towards  $j$ .”

## 2 First Steps

Fix a finite planar graph  $G = (V, E)$ , as well as an orientation  $K$  of the edges thereof to be chosen later. Let us moreover denote by  $A$  the weighted adjacency

matrix associated to  $K$ , as defined in the previous section. We begin by noting that if  $|V|$  is odd, then we must have that  $\mathcal{D} = \emptyset$ , as any perfect matching of  $G$  defines a partition of the set  $V$  into pairs. Subsequently, in this case Kasteleyn's theorem reduces to the claim that  $\det A = 0$ . This is an elementary consequence of skew-symmetry and is left to the reader as an exercise.

It remains to address the interesting case when  $|V|$  is even. To be explicit, let us assume that  $|V| = 2n$ , and let us moreover fix an enumeration  $1, 2, \dots, 2n$  of the members of  $V$ . As  $A$  is skew-symmetric, we can consider its *Pfaffian*:

$$\text{pf}(A) = \frac{1}{2^n n!} \sum_{\sigma \in S_{2n}} \text{sgn}(\sigma) a_{\sigma(1)\sigma(2)} \cdots a_{\sigma(2n-1)\sigma(2n)}.$$

In particular, if we recall that  $\text{pf}(A)$  is characterized by the property that  $\text{pf}(A)^2 = \det A$ , then we see that Kasteleyn's theorem is equivalent to choosing  $K$  so that  $Z = |\text{pf}(A)|$ . As a first step in pursuit of this, we provide a useful reformulation of  $\text{pf}(A)$  which renders its connection to perfect matchings of  $G$  more transparent:

**Proposition.** *The value of  $\text{sgn}(\sigma) a_{\sigma(1)\sigma(2)} \cdots a_{\sigma(2n-1)\sigma(2n)}$  is entirely determined by the “matching” of  $\{1, \dots, 2n\}$  into unordered pairs  $\{\sigma(1), \sigma(2)\}, \dots, \{\sigma(2n-1), \sigma(2n)\}$ . In particular, if we let*

$$\Pi = \{\pi = \{\{i_1, j_1\}, \dots, \{i_n, j_n\}\}\}$$

*denote the set of such matchings, then we have that*

$$\text{pf}(A) = \sum_{\pi \in \Pi} \text{sgn}(\bar{\sigma}) a_{i_1 j_1} \cdots a_{i_n j_n}$$

*where  $\bar{\sigma}: (1, \dots, 2n) \mapsto (i_1, j_1, \dots, i_n, j_n)$ .*

*Proof.* Induct on the order  $2n$  of the permutation  $\sigma$  and use skew-symmetry of  $A$ . The expression for  $\text{pf}(A)$  follows from the fact that there are  $2^n n!$  permutations which induce any given matching  $\pi = \{\{i_1, j_1\}, \dots, \{i_n, j_n\}\}$ :  $n!$  ways to assign each pair  $\{\sigma(k), \sigma(k+1)\}$  to a pair  $\{i_\ell, j_\ell\}$ , plus an additional factor of  $2^n$  since it doesn't matter which of  $\sigma(k)$  or  $\sigma(k+1)$  gets assigned to which number in the pair.  $\square$

The usefulness of this reformulation of the Pfaffian of  $A$  stems entirely from the observation that to each matching  $\pi = \{\{i_1, j_1\}, \dots, \{i_n, j_n\}\}$  of the numbers  $1, \dots, 2n$  into unordered pairs, there is an associated collection of dimer configurations of  $G$ ; namely, the set  $\mathcal{D}_\pi$  of all  $\{e_1, \dots, e_n\}$ , where  $e_\ell$  is any choice of edge between  $i_\ell$  and  $j_\ell$ . With this correspondence  $\pi \leftrightarrow \mathcal{D}_\pi$  in mind, we find that

$$\begin{aligned} \text{pf}(A) &= \sum_{\pi \in \Pi} \text{sgn}(\bar{\sigma}) \prod_{\ell=1}^n \sum_e \varepsilon_{i_\ell j_\ell}(e) \nu(e) \\ &= \sum_{\pi \in \Pi} \text{sgn}(\bar{\sigma}) \sum_{D \in \mathcal{D}_\pi} \prod_{\ell=1}^n \varepsilon_{i_\ell j_\ell}(e) \nu(e) = \sum_{D \in \mathcal{D}} \varepsilon(D) \nu(D), \end{aligned}$$

where  $\varepsilon(D)$  denotes the “sign” factor  $\varepsilon(D) := \text{sgn}(\bar{\sigma}) \prod_{\ell=1}^n \varepsilon_{i_\ell j_\ell}(e_\ell) \in \{\pm 1\}$ . Notice that up to this factor, this is exactly the partition function  $Z$ . We therefore see that if we can choose our orientation  $K$  so that this sign term  $\varepsilon(D)$  is independent of  $D$ , then it will indeed be the case that  $Z = |\text{pf}(A)|$ . Such an orientation shall be called a *Pfaffian orientation*.

### 3 Constructing the Pfaffian Orientation

One way we can ensure that  $\varepsilon(D)$  is independent of  $D$  is by endeavouring to choose  $K$  so that for any two distinct dimer configurations  $D, D'$  of  $G$ , we have that  $\varepsilon(D)\varepsilon(D') = 1$ . The following result gives a characterization of how  $D$  and  $D'$  relate to one another that will be useful for achieving this.

**Lemma 1.** *Given distinct dimer configurations  $D, D' \in \mathcal{D}$  of  $G$ , their symmetric difference  $D\Delta D'$  is given by the disjoint union of simple closed curves of even length in  $G$ , the edges of which alternate between  $D$  and  $D'$ .*

*Proof.* Consider an edge  $e_0$  in the symmetric difference  $D\Delta D'$ . It is sufficient to show that  $e_0$  lies on a cycle of even length in  $G$  whose edges alternate between  $D$  and  $D'$ ; perfection of the matchings  $D$  and  $D'$  ensure that this can be true for only one such cycle.

To that end, we begin by noting that the edge  $e_0$  must lie in one of  $D$  or  $D'$ , but not both; without loss of generality, let us suppose  $e_0 \in D$ . Denote by  $v_0$  and  $w_0$  the endpoints of  $e_0$  and note that because  $D'$  is a perfect matching, we can find a unique edge  $e'_0 \in D'$  with  $v_0$  as an endpoint. Notice that because  $e_0 \notin D'$ , the edges  $e_0$  and  $e'_0$  must be distinct. If we then denote by  $w'_0$  the other endpoint of  $e'_0$ , we see that if  $w'_0 = w_0$  then we are done. Otherwise, we can use perfection of  $D$  to find an edge  $e_1$  with  $v_1 := w'_0$  as an endpoint. In this case, perfection of  $D$  also guarantees that the other endpoint  $w_1$  of  $e_1$  must be distinct from  $w_0$ , so let us additionally appeal to perfection of  $D'$  to find  $e'_1 \in D'$  having  $v'_1 = w_1$  as one of its endpoints and  $w'_1$  the other.

Again, we note that if  $w'_1 = w_0$  then we are done. If not, then we continue this process. By finiteness of  $G$  it must terminate after some number of steps  $m$ , whence the edges  $e_0, e'_0, e_1, e'_1, \dots, e_m, e'_m$  form the desired cycle.  $\square$

The above lemma provides us with a decomposition of  $D\Delta D'$  into nice curves  $C_1, \dots, C_m$ . Moreover, since we’ve fixed a labelling for the vertices of  $G$ , notice that each such curve corresponds in a natural way to a cyclic permutation  $\sigma_{C_\ell} \in S_{2n}$ : namely, if  $i_1$  is any vertex in  $C_\ell$  and  $i_2, \dots, i_k$  are the others when we move along  $C_\ell$  in the clockwise direction starting at  $i_1$ , then we set  $\sigma_{C_\ell} = (i_1 i_2 \dots i_k)$ . Geometrically, the permutation  $\sigma_{C_\ell}$  traverses the curve  $C_\ell$  clockwise. To see why these permutations are useful, let us take a more detailed look at the product  $\varepsilon(D)\varepsilon(D')$ . In particular, let us say that  $D$  consists of edges  $e_1, \dots, e_n$  matching vertices  $i_\ell$  and  $j_\ell$ , and that  $D'$  consists of edges  $e'_1, \dots, e'_n$  matching vertices  $i'_\ell$

to  $j'_\ell$ . Then

$$\varepsilon(D)\varepsilon(D') = \text{sgn}(\sigma_D) \prod_{\ell=1}^n \varepsilon_{i_\ell j_\ell}(e_\ell) \cdot \text{sgn}(\sigma_{D'}) \prod_{\ell=1}^n \varepsilon_{i'_\ell j'_\ell}(e'_\ell)$$

where  $\sigma_D$  and  $\sigma_{D'}$  denote the permutations  $(1, \dots, 2n) \mapsto (i_1, j_1, \dots, i_n, j_n)$  and  $(1, \dots, 2n) \mapsto (i'_1, j'_1, \dots, i'_n, j'_n)$ , respectively. Let us moreover simplify this to

$$\varepsilon(D)\varepsilon(D') = \text{sgn}(\sigma_D) \text{sgn}(\sigma_{D'}) \prod_{\ell=1}^n \varepsilon_{i_\ell j_\ell}(e_\ell) \varepsilon_{i'_\ell j'_\ell}(e'_\ell)$$

As we shall see in the next talk, the factor  $\prod_{\ell=1}^n \varepsilon_{i_\ell j_\ell}(e_\ell) \varepsilon_{i'_\ell j'_\ell}(e'_\ell)$  is easily computable in terms of the curves  $C_1, \dots, C_m$ . The other factor,  $\text{sgn}(\sigma_D) \text{sgn}(\sigma_{D'})$ , is unfortunately a little more tricky. The saving grace that will ultimately allow us to deal with it is the proposition of the previous section, which says that we needn't have chosen  $\sigma_D$  and  $\sigma_{D'}$  as we have here; rather, there are a good many other choices that we could've used instead. A reasonable question to ask, then, is whether we can choose  $\sigma_D$  and  $\sigma_{D'}$  “nicely” so that the product  $\text{sgn}(\sigma_D) \text{sgn}(\sigma_{D'})$  reduces to something elementary. This is precisely where the permutations  $\sigma_{C_1}, \dots, \sigma_{C_m}$  come in:

**Lemma 2.** *It is possible to choose  $\sigma_D$  and  $\sigma_{D'}$  so that  $\sigma_{D'} \circ \sigma_D^{-1} = \prod_{\ell=1}^m \sigma_{C_\ell}$ . In particular,  $\text{sgn}(\sigma_D) \text{sgn}(\sigma_{D'}) = (-1)^m$ .*

*Proof.* The idea is just to start with any choice of  $\sigma_D$  and then construct a suitable choice of  $\sigma_{D'}$  step by step. In particular, let us take the “standard” representative  $\sigma_D: (1, \dots, 2n) \mapsto (i_1, j_1, \dots, i_n, j_n)$ , so that  $\sigma_D^{-1}: (i_1, j_1, \dots, i_n, j_n) \mapsto (1, \dots, 2n)$ . Consider the pair of vertices  $(i_1, j_1)$ . We know that  $e_1 \in D$  matches these vertices. If  $e_1$  is also in  $D'$ , then  $e_1$  does *not* lie in the symmetric difference; the permutation  $\prod_{\ell=1}^m \sigma_{C_\ell}$  thus fixes these vertices, so we want to choose  $\sigma_{D'}$  so that  $\sigma_{D'} \circ \sigma_D^{-1}$  does as well. This is easy enough: we just set  $\sigma_{D'}(1) = i_1$  and  $\sigma_{D'}(2) = j_1$ . On the other hand, if  $e_1$  does not belong to  $D'$ , then  $e_1$  lies on some curve  $C_\ell$  in the symmetric difference  $D \Delta D'$ . Because the edges of this curve are alternating, we (without loss of generality) have  $j_1 = i'_k$  for some  $k \in \{1, \dots, n\}$ . Again, to mimic the behavior of  $\prod_{\ell=1}^m \sigma_{C_\ell}$ , we set  $\sigma_{D'}(1) = i'_k$  and  $\sigma_{D'}(2) = j'_k$ . Now consider the pair  $(i_2, j_2)$  and do the same thing to get a definition for  $\sigma_{D'}(3)$  and  $\sigma_{D'}(4)$ , and continue until  $\sigma_{D'}$  is defined completely.

The final remark follows because we have  $\text{sgn}(\sigma_D) \text{sgn}(\sigma_{D'}) = \text{sgn}(\sigma_{D'} \circ \sigma_D^{-1})$ , and each  $\sigma_{C_\ell}$  has sign  $-1$  as a cyclic permutation of even length.  $\square$